Problem 5. Consider a regression model $Y_i = \beta x_i + \varepsilon_i$ for $i = 1, \ldots, n$, where ε_i are independent and $N(0, \sigma^2)$. It is a version of a simple regression model with zero α, namely, when given $X = 0$, mean of Y is zero. Points x_i are assumed to be constants (not random variables).

See also Problem 6.5-2

Derive the maximum likelihood estimators, $\hat{\beta}$ and $\hat{\sigma}^2$, of β and σ^2 under this model. Do not use the known formulas for a simple regression problem. Show all the steps. What is the distribution of Y_i? Of $\hat{\beta}$? ⊛

$\varepsilon_i \sim N(0, \sigma^2)$, so $\boxed{Y_i \sim N(\beta x_i, \sigma^2)}$ [x_i's assumed to be constants]

Maximum likelihood function is

$$L(\sigma^2, \beta, y_1 \ldots y_n) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\sum_{i=1}^{n}(y_i - \beta x_i)^2}{2\sigma^2}\right)$$

$$\log L = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{\sum_i (y_i - \beta x_i)^2}{2\sigma^2}$$

$$(\log L)'_\beta = \frac{\sum 2(y_i - \beta x_i)\cdot x_i}{2\sigma^2} = 0 \Rightarrow \boxed{\hat{\beta} = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} x_i^2}}$$

$$(\log L)'_{\sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum_i (y_i - \hat{\beta} x_i)^2}{2(\sigma^2)^2} = 0$$

$$\boxed{\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\beta} x_i)^2}$$

using second derivative test or looking at the sign of derivative function we can check that these points actually give maximum of $L(\sigma^2, \beta)$

$$\boxed{\hat{\beta} \sim N\left(\frac{\sum_{i=1}^{n} \beta x_i^2}{\sum_{i=1}^{n} x_i^2}, \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}\right)}$$

as x_i's are constants, and y_i's have distribution ⊛

Problem 6. Is my coin fair? You flip a coin 200 times. It comes up head 85 many times and tails 115 many times. Using the Chi-Square test, do you accept the hypothesis that your coin is fair at the 5% significance level? Report an interval for the p-value.

2 classes (head, tail) $k=2$

$n=200$

$$q_{2-1} = \sum_{i=1}^{2} \frac{\left(y_i - n\cdot\frac{1}{2}\right)^2}{n\cdot\frac{1}{2}} = \frac{(85-100)^2}{100} + \frac{(115-100)^2}{100}$$

$$= \frac{2\cdot 15^2}{100} = 4.5$$

Critical region $\{q_{k-1} > \chi^2_\alpha(k-1)\}$

$\chi^2_{0.05}(1) \approx 3.8$

Since $4.5 > 3.8$, we reject H_0
(coin is <u>not</u> fair)

Problem 7. You have a coin that shows head (i.e. $X = 1$) with probability $\theta \in [0, 1]$ and tail (i.e. $X = 0$) with probability $1 - \theta$. You do not know the parameter θ, but you assume that θ is distributed according to the prior distribution

$$f_\theta(t) = ct, \text{ if } 0 \le t \le 1 \quad (*)$$

and 0 otherwise. Assume you observe tail in the next toss, what is the posterior distribution of θ? How would you estimate success probability then? (Hint: what is the MAP estimate of θ?)

(X is discrete
θ is continuous)
$t = \theta$ in $(*)$

Bayes rule:

$$f_{\theta|X}(\theta|x) = \frac{\mathbb{P}(X=x|\theta=\theta)\cdot f(\theta)}{\int \mathbb{P}(X=x|\theta=\theta')\cdot f(\theta')\,d\theta'} =$$

Since $x = 0$ (tail)

$$= \frac{(1-\theta)\cdot c\theta}{\int_0^1 (1-\theta)\cdot c\theta\,d\theta} = \frac{(1-\theta)\,\theta}{\int_0^1 (1-\theta)\,\theta\,d\theta}$$

$$\int_0^1 (1-\theta)\cdot\theta\,d\theta = \int_0^1 \theta - \theta^2\,d\theta = \frac{\theta^2}{2} - \frac{\theta^3}{3}\Big|_0^1 = \frac{1}{6}$$

Posterior distribution $\boxed{f_{\theta|X}(\theta|1) = 6(1-\theta)\,\theta \text{ if } \theta \in [0,1] \text{ and } 0 \text{ otherwise.}}$

$$\text{MAP} = \arg\max_\theta f_{\theta|X}(\theta|1)$$

$((1-\theta)\theta)' = 1 - 2\theta = 0$ $\quad \theta = \frac{1}{2}$, and it is maximum since $((1-\theta)\theta)'' = -2$

$\boxed{\text{MAP } \hat{\theta} = \frac{1}{2}}$

Problem 8. Let $X_1, \ldots, X_n$ be a random sample from the distribution with the density function $f(x;\theta) = 0.5e^{-|x-\theta|}$, $-\infty < x < \infty$ and $-\infty < \theta < \infty$. Find maximum likelihood estimator θ.

typo was here → See also the hint in the exercise 6.4-~~8~~ 5 of the textbook.

Maximum likelihood function

$$L(\theta, x_1, \ldots x_n) = \prod_{i=1}^{n} (0.5) \exp(-|x_i - \theta|) =$$

$$= \frac{1}{2^n} \cdot \exp\left(-\sum_{i=1}^{n} |x_i - \theta|\right)$$

L is not differentiable, so we need to find best θ directly: to minimize $\sum_{i=1}^{n} |x_i - \theta|$, $\boxed{\hat{\theta} = \text{median}(X_1 \ldots X_n)}$

Suggestion: Do also 6.4-5 a) and b) as more standard maximum likelihood exercises

Problem 9. A sample size of $n = 100$ is taken from the production of light switches. A quality control engineer discovered that 10 switches does not meet the quality standard.

a) Suppose that another 500 switches are to be produces using the same technology. What are minimal and maximal amounts of defective switches should we expect with 95% confidence?

Hint: Use confidence interval for the defect rate of the production (=proportion of bad switches) to answer this question.

Confidence interval for p - proportion of defective switches:

$$p \in \left[\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right] = \left[0.1 - z_{0.05}\sqrt{\frac{0.1\cdot 0.9}{500}};\ 0.1 + z_{0.05}\sqrt{\frac{0.1\cdot 0.9}{500}}\right]$$

prior estimate for p ($\hat{p} = 10/100$)

$$\Leftrightarrow \left[0.1 - 1.96\,\frac{3}{\sqrt{5}\cdot 100};\ 0.1 + 1.96\,\frac{3}{\sqrt{5}\cdot 100}\right]$$

So, from 500 switches, there are at most

$$\left\lfloor 500\cdot\left(0.1 + 1.96\cdot\frac{3}{\sqrt{5}\cdot 100}\right)\right\rfloor = \left\lfloor 50 + 1.96\cdot 3\cdot\sqrt{5}\right\rfloor \approx 63$$

round down

defective switches

and at least

$$\left\lceil 500\left(0.1 - 1.96\,\frac{3}{\sqrt{5}\cdot 100}\right)\right\rceil = \left\lceil 50 - 1.96\cdot 3\sqrt{5}\right\rceil \approx 37$$

round up

defective switches.

b) Estimate the confidence that there are at most 60 defective switches:

$$\mathbb{P}\left(\frac{y}{n} \leq \frac{60}{500}\right) = \mathbb{P}\left(\frac{\frac{y}{n} - \hat{p}}{\sqrt{\hat{p}(1-\hat{p})/n}} \leq \frac{\frac{6}{50} - 0.1}{\sqrt{0.1 \cdot 0.9 / 500}}\right)$$

$$= \mathbb{P}\left(\underset{\sim N(0,1)}{Z} \leq \frac{100 \cdot \sqrt{5} \cdot 1}{3 \cdot 50}\right) = \mathbb{P}\left(Z \leq \frac{10}{3\underset{\approx 2.5}{\sqrt{5}}}\right) \approx$$

$$\mathbb{P}\left(Z \leq \frac{4}{3}\right) \approx 0.5082$$

b) ~~Estimate the confidence (=probability) that among the next 500 switches will be at most 60 defective switches?~~

Problem 10. On some magical island, there were elves, dwarfs, orcs and goblins. A researcher arrives to the island and decides to interview all four spices about their history and traditions. She invites 10 representatives of each folk for an interview. However, she soon finds out that at least some of the island inhabitants are quite mean and aggressive and don't interview well (whereas the rest are quite nice and friendly). So, the researcher wonders: are there more and less aggressive spices on the island? She records the data she got so far with in the following table:

	nice	not nice	
elves:	3	7	10
dwarfs:	5	5	10
orcs:	4	6	10
goblins:	8	2	10
	20	20	

How can she test on the significance level 10% a hypothesis that being nice is a spices independent feature?

Specify what test you will use. Show all the steps. Make a conclusion (whether the researcher should accept or reject her hypothesis about independence).

We can use χ^2-test for the independence of 2 factors.

$k=2,\ h=4 \qquad (k-1)(h-1)=3$

$$Q_3 = \sum_{j=1}^{h}\sum_{i=1}^{k} \frac{\left(y_{ij} - n\frac{y_{i\cdot}}{n}\cdot\frac{y_{\cdot j}}{n}\right)^2}{n\frac{y_{i\cdot}}{n}\cdot\frac{y_{\cdot j}}{n}}$$

$$n\cdot\frac{y_{i\cdot}}{n}\,\frac{y_{\cdot j}}{n} = \frac{10}{40}\cdot\frac{20}{40}\cdot 40 = 5 \quad \text{for all } i,j$$

So, $$Q_3 = \frac{(3-5)^2}{5} + \frac{(7-5)^2}{5} + 0 + 0 + \left(\frac{4-5}{5}\right)^2 + \frac{(6-5)^2}{5} + \frac{(8-5)^2}{5}$$

$$+\frac{(2-5)^2}{5} = \frac{4+4+1+1+9+9}{5} = \boxed{\frac{28}{5}}$$

$\chi^2_{0.1}(3) = 6.25$ $\qquad \frac{28}{5} < 6 < 6.25 \Rightarrow \boxed{\text{accept } H_0}$

Critical region $\{Q_3 > \chi^2_{0.1}(3)\}$